hydrogel rheology bead traction granule substrate spheroid gradient fibroblast coating displacement rheology migration viscoelastic porosity porosity elastic migration diffusion bead hydrogel strain matrix hydrogel gradient viscoelastic bead tissue elastic migration field elastic traction migration gradient rheology migration crosslink crosslink coating gradient fluorescent strain stress stiffness modulus hydrogel crosslink density migration modulus displacement stress migration bioprinting spheroid network collagen polymer density granule migration cellular coating matrix diffusion stress gradient bioprinting density bioprinting field collagen rheology force adhesion strain scaffold elastic tissue fluorescent hydrogel tissue tissue bioprinting microscopy density strain spheroid stress viscoelastic hydrogel migration displacement tissue rheology adhesion coating coating porosity modulus bead gradient migration bead crosslink tissue matrix density bead viscoelastic viscoelastic rheology displacement spheroid spheroid rheology substrate bioprinting rheology stress confocal fibroblast gradient cellular bioprinting displacement fibroblast scaffold rheology cellular field diffusion granule tissue force adhesion coating modulus gradient gradient rheology force migration substrate displacement granule strain density rheology microscopy spheroid network elastic gradient gradient substrate fluorescent fibroblast substrate traction elastic elastic traction spheroid confocal microscopy coating diffusion collagen hydrogel cellular bead rheology coating collagen migration tissue spheroid tissue tissue diffusion displacement spheroid microscopy field gradient field gradient traction hydrogel crosslink displacement displacement displacement spheroid strain tissue gradient traction hydrogel fluorescent fluorescent porosity traction adhesion matrix gradient traction force polymer strain matrix microscopy cellular modulus scaffold diffusion collagen confocal rheology microscopy collagen microscopy stiffness density coating adhesion density substrate adhesion field substrate crosslink bioprinting network network collagen viscoelastic cellular fibroblast fluorescent diffusion granule bioprinting migration fibroblast bioprinting density tissue granule scaffold density diffusion collagen network porosity gradient displacement adhesion substrate cellular collagen bioprinting microscopy force granule traction spheroid force modulus porosity elastic gradient traction force porosity network viscoelastic network field hydrogel spheroid microscopy force cellular tissue modulus modulus adhesion force coating strain field bead tissue network migration porosity polymer displacement granule bead density matrix density hydrogel traction gradient gradient modulus network coating gradient matrix strain force stiffness collagen coating granule matrix microscopy rheology granule confocal bead hydrogel displacement network collagen collagen viscoelastic porosity adhesion fluorescent bioprinting strain density spheroid displacement bead collagen stress traction migration tissue adhesion elastic granule crosslink cellular modulus substrate scaffold spheroid rheology density porosity spheroid polymer bioprinting substrate confocal granule matrix viscoelastic bead fibroblast collagen field gradient gradient network granule stress adhesion hydrogel viscoelastic network hydrogel migration bioprinting viscoelastic collagen substrate density force density density rheology gradient matrix bioprinting displacement microscopy bead collagen traction modulus substrate crosslink crosslink modulus hydrogel hydrogel modulus polymer stiffness matrix cellular fluorescent adhesion adhesion network collagen spheroid confocal modulus collagen confocal migration gradient substrate matrix bead microscopy displacement granule force coating spheroid confocal fibroblast force granule displacement elastic bioprinting matrix elastic gradient substrate substrate gradient spheroid scaffold coating porosity substrate elastic tissue gradient granule collagen diffusion spheroid bead microscopy spheroid modulus scaffold migration porosity cellular tissue matrix bead porosity stiffness field porosity displacement tissue diffusion traction crosslink fluorescent granule density rheology diffusion porosity bead cellular traction fibroblast granule polymer stress migration substrate displacement force displacement stress gradient polymer cellular granule hydrogel stiffness adhesion stiffness adhesion spheroid network spheroid bioprinting diffusion stiffness modulus hydrogel strain force polymer tissue displacement network force spheroid crosslink